MADHESWARAN S

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EXECUTIVE SUMMARY

Electronics and Communication Engineering student skilled in Embedded Systems, IoT, and VLSI, with hands-on experience building ESP32- and sensor-based academic projects. The entry-level technology roles I am targeting require strong problem-solving, quick learning, and the ability to translate embedded-systems concepts into working prototypes — strengths I have demonstrated through projects, workshops, and paper presentations. I am confident I can contribute effectively to a collaborative engineering team from day one.

TECHNICAL SKILLS

- Embedded Systems: Microcontrollers, Sensor-based Systems
- IoT: ESP32, IoT-based Communication
- Electronics: Accident Detection Systems, Signal Control
- Areas of Interest: Embedded Systems, Internet of Things (IoT), VLSI

SOFT SKILLS

- Problem Solving
- Quick Learning
- Team Collaboration
- Technical Presentation
- Interpersonal Skills

PROFESSIONAL EXPERIENCE

AI-Driven Urban Traffic System with Priority Routing for Ambulance — Academic Project
Role: Project Contributor

Responsibilities:

- Developed an intelligent traffic management system focused on emergency vehicle priority.
- Worked on signal control mechanisms for efficient traffic management.
- Implemented IoT-based communication for emergency vehicle coordination.
- Used ESP32 as part of the project implementation.
- Applied embedded-system concepts to support priority routing for ambulances.

Impact:

- Developed an academic prototype focused on improving emergency vehicle movement through intelligent signal control.
- Integrated IoT communication and ESP32-based technology into the solution.

Smart Helmet — Academic Project
Role: Project Contributor

Responsibilities:

- Designed a smart helmet concept focused on rider safety.
- Developed accident detection functionality using sensors.
- Worked with microcontroller technology for system implementation.
- Integrated sensor-based inputs for safety monitoring.
- Applied embedded-system concepts to develop the project.

Impact:

- Developed an academic safety solution combining accident detection and rider protection features.
- Demonstrated practical application of sensors and microcontroller technology.

EDUCATION

B.E. Electronics and Communication Engineering

M. Kumarasamy College of Engineering, Anna University | 2026

COURSES

- NPTEL – Introduction to Industry 4.0 and Industrial Internet of Things

AWARDS & ACCOMPLISHMENTS

- Participated in a Cyber Security Workshop conducted by Kongunadu College of Engineering and Technology.
- Presented a paper on Smart Helmet at Sengunthar College of Engineering.

LANGUAGES

- English – Working Proficiency
- Tamil – Fluent